

EyeDiseaseNet: An Explainable and Resource-Efficient Self-Attention Hybrid Network for Multi-Class Ocular Disease Classification

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Abstract

Ocular diseases including cataracts, uveitis, conjunctivitis, and eyelid disorders, are primary causes of global visual impairment. Early identification is essential to prevent permanent vision loss, yet resource-constrained areas frequently lack accessible ophthalmic screening tools. Although deep learning architectures show promise in automated diagnostics, existing systems often struggle to distinguish visually similar conditions, demand excessive computational resources, and operate as uninterpretable black boxes. To address these limitations, we present EyeDiseaseNet, a multi-stream hybrid deep learning architecture designed for multi-class ocular disease classification. Our framework implements a robust image preprocessing pipeline combining Contrast Limited Adaptive Histogram Equalization (CLAHE) and Non-Local Means Denoising to eliminate noise and correct illumination artifacts. The model combines global and local structural features by fusing VGG19 and MobileNetV2 representations, which are further augmented with a lightweight self-attention mechanism to isolate fine pathological patterns. Furthermore, we integrate Gradient-weighted Class Activation Mapping (Grad-CAM) to generate spatial attention heatmaps, providing visual explanations for each diagnosis. Experimentally evaluated on a balanced multi-class dataset of 6,894 anterior segment eye images, EyeDiseaseNet achieved an overall classification accuracy of 99.13% and a Cohen's Kappa score of 0.9889, markedly outperforming standalone baseline models. The model maintained computational efficiency with a training duration of 554.46 seconds and an inference time of 12.14 seconds. These results demonstrate that EyeDiseaseNet provides an accurate, explainable, and resource-efficient solution highly suitable for real-world clinical decision support systems.

1. Introduction

Ocular diseases cause vision problems for many people worldwide. These conditions include cataracts, uveitis, conjunctivitis, and eyelid disorders [13]. The World Health Organization states that over 2.2 billion people suffer from vision impairment globally [20]. Early detection can prevent or treat many of these cases [5]. However, rural and low-resource areas lack eye doctors and diagnostic tools [3]. Patients often face long travel distances and high clinical costs. This shortage leads to delayed diagnoses and permanent vision loss [3]. Automated screening systems can provide quick assessments in remote clinics.

Recently, deep learning has improved medical image analysis significantly. Researchers use Convolutional Neural Networks (CNNs) to classify different eye diseases [11, 8]. Yet, existing methods face several technical issues. First, poor preprocessing creates noisy or low-contrast images. Ocular images often contain artifacts from poor lighting and camera movement. Second, models lack explainability [18]. Clinicians do not trust automated decisions without knowing the underlying reasoning. Third, models often misclassify visually similar conditions. For example, separating uveitis from conjunctivitis is difficult for standard networks [19, 9]. Finally, high-accuracy networks require immense computational power [21]. This requirement prevents real-time deployment on small, low-cost devices.

This paper introduces EyeDiseaseNet to solve these challenges. EyeDiseaseNet is an automated framework for multiclass eye disease classification. The system combines features from two distinct architectures: VGG19 [17] and

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MobileNetV2 [7]. We add a self-attention mechanism to this combined network [6]. The self-attention module helps the model focus on tiny, important image details. This approach keeps the model lightweight while boosting accuracy. Our preprocessing pipeline uses Contrast Limited Adaptive Histogram Equalization (CLAHE) [16] and Non-Local Means Denoising [2]. CLAHE improves local contrast. The denoising algorithm removes noise without blurring essential structural edges. We also integrate Gradient-weighted Class Activation Mapping (Grad-CAM) into the model [1]. Grad-CAM generates heatmaps that highlight the eye regions affecting the classification choices. This visual evidence helps doctors verify and trust the system output.

The core contributions of this research are as follows:

- We create a feature-fusion network combining VGG19 and MobileNetV2. We add a self-attention mechanism to reduce classification errors.
- We implement a preprocessing pipeline using CLAHE and Non-Local Means Denoising. This pipeline enhances image contrast and removes noise.
- We achieve high multiclass classification performance across five categories: eyelid disorders, normal eyes, cataracts, uveitis, and conjunctivitis.
- We incorporate Grad-CAM to visualize affected eye areas. This addition provides transparency for clinical use.

2. Related Work

Many researchers study deep neural networks for medical image processing [11]. Erdin and Patel created a convolutional neural network (CNN) with nine layers to detect eye conditions [8]. They checked for trachoma, cataracts, and conjunctivitis. Their model reached an accuracy of 88.36%. Rashid et al. used a deep CNN with 16 layers to classify external eye diseases [15]. Their network used dropout layers to stop overfitting and achieved 98.48% accuracy. Tashkandi used retinal images to predict eye diseases with deep learning [18]. Abd El-Ghany et al. designed a two-dimensional CNN to detect retinal asymmetries [1]. They included Gradient-weighted Class Activation Mapping (Grad-CAM) to show high-accuracy classifications.

Other studies look at single eye diseases like cataracts. Nguyen and Lin developed a hybrid CNN system for cataract screening [12]. They divided images into four parts and used a voting method to get 97.12% accuracy. Lahari et al. built a small deep learning framework named CSDNet [10]. This framework had only 175,000 parameters. It achieved 97.3% accuracy for cataract detection. Umamaheswari et al. designed a system to diagnose conjunctivitis using image segments [19]. Ghadiri and Thomas reviewed neural networks that forecast the causes of uveitis [9]. They noted that data scarcity and complex models make clinical usage difficult.

Image preparation and feature handling are also important in this area. Zannah et al. built a machine learning model using Bayesian optimization [21]. They extracted deep features using networks like EfficientNet and reduced the dimensions with Principal Component Analysis. Their system achieved 95.33% accuracy on fundus images [22]. Chakraborty et al. created a dual attention-aided DenseNet-121 architecture [6]. They used this network to classify glaucoma from fundus pictures. Ahmad et al. analyzed how image enhancement affects medical image segmentation [2]. They compared Contrast-Limited Adaptive Histogram Equalization (CLAHE), histogram equalization, and anisotropic diffusion. Their tests showed that CLAHE improves model results. Pannu et al. improved glaucoma detection by extracting cup-to-disc ratio features and neuro-retinal rim data [14].

2.1. Research Gaps

Current systems have several limitations. Many models only identify one type of eye disease, such as cataracts or glaucoma. They fail when processing multiple external and internal eye conditions together. In addition, existing frameworks often lack strong image enhancement steps. Noise and low contrast lower their accuracy in real-world clinics. Many deep learning architectures also require massive computational power. This requirement makes it hard to run them on low-cost portable hardware. Finally, most models act like black boxes. They do not show doctors why a decision was made, which reduces medical trust.

2.2. Addressing the Research Gaps

Our proposed model, EyeDiseaseNet, solves the issues found in previous studies. First, many older models only detect a single eye condition [10]. They fail when testing multi-disease patient data. We solve this by training our model

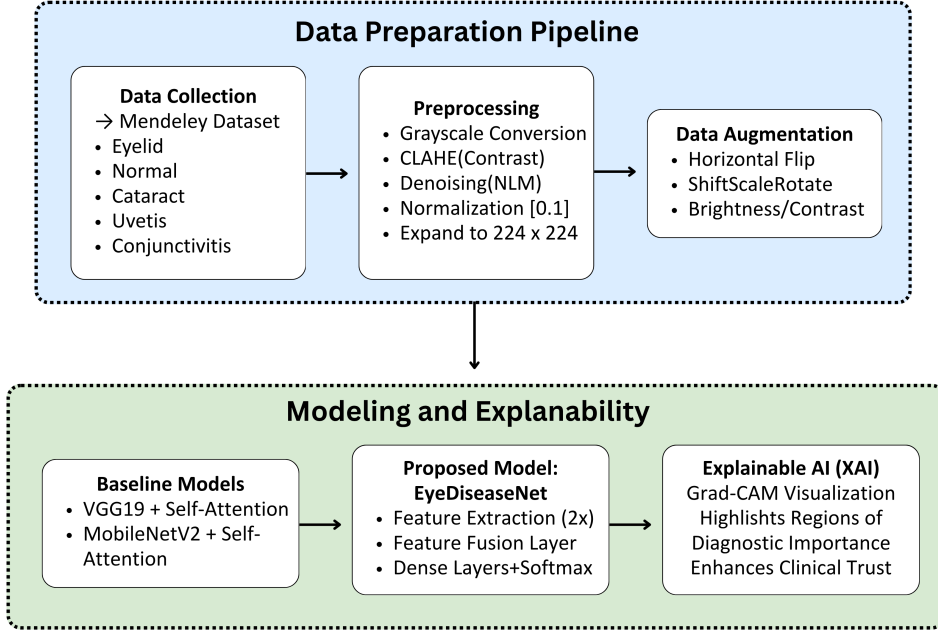


Figure 1: Workflow diagram.

on five distinct categories together [4]. These categories include eyelid disorders, cataracts, uveitis, conjunctivitis, and normal eyes. This setup makes our framework practical for general eye clinics.

Second, poor image quality often causes high misclassification rates [19]. Standard networks struggle to separate visually similar conditions like uveitis and conjunctivitis [9]. We fix this issue by creating a solid preprocessing step. This step combines Contrast Limited Adaptive Histogram Equalization (CLAHE) and Non-Local Means Denoising [2]. CLAHE increases the local details of the eye. The denoising step removes grain without losing critical tissue edges. This clean input helps our system separate similar conditions easily.

Third, high accuracy usually requires enormous computer hardware [21]. This high requirement stops real-time deployment in remote areas. We solve this by designing a hybrid network. We fuse VGG19 and MobileNetV2 features. Then, we add a lightweight self-attention mechanism [6]. This attention step focuses only on the most important eye regions. It gives high predictive accuracy without needing heavy computing resources.

Finally, most deep learning models operate as black boxes [18]. Doctors do not trust decisions from hidden algorithms. We solve this lack of transparency by integrating Gradient-weighted Class Activation Mapping (Grad-CAM) [1]. Grad-CAM creates visual maps that highlight exactly where the model looks. Doctors can see the exact features that led to the final diagnosis. This visual evidence bridges the gap between machine learning and clinical trust.

3. Methodology

We developed a structured deep learning pipeline to classify various ocular conditions. The complete workflow of our method contains data acquisition, preprocessing, image augmentation, network design, and explainability, as shown in Figure 1.

3.1. Data Collection

We evaluated our approach using a publicly available dataset of eye photos compiled by Bitto et al. [4]. This collection contains images focusing on the anterior segments of human eyes. The repository partitions the samples into five distinct diagnostic categories: Eyelid, Conjunctivitis, Uveitis, Cataract, and Normal. Specifically, the original database includes 357 images of Conjunctivitis, 223 images of Uveitis, 544 images of Cataract, and 649 images of

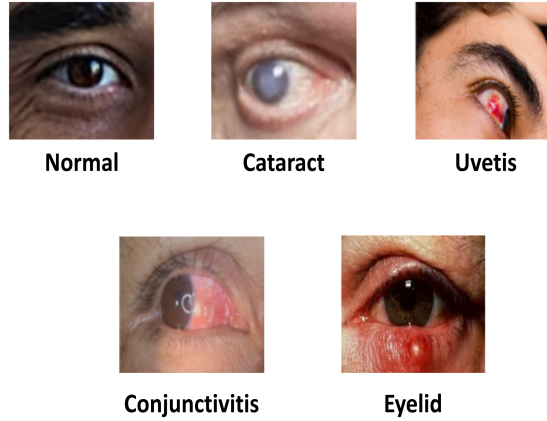


Figure 2: Sample images from the dataset.

Table 1
Image Preprocessing Steps and Parameters

Step	Parameters
Grayscale Conversion	<code>cv2.IMREAD_GRAYSCALE</code>
Contrast Enhancement	CLAHE: <code>clipLimit=2.0</code> , <code>tileGridSize=(8,8)</code>
Noise Reduction	FastNlMeansDenoising: <code>h=10</code>
Normalization	Scale pixels to $[0, 1]$ by dividing by 255
Resizing	Resize to 224×224
Channel Expansion	Convert GRAY to RGB using <code>cv2.cvtColor</code>

Normal eyes. These samples provide a diverse range of pathological features. This diversity establishes a reliable foundation to train and validate deep learning systems for automated disease identification. Figure 2 illustrates representative examples from this dataset.

3.2. Preprocessing

We implemented a systematic preprocessing pipeline to enhance the quality and uniformity of the input photographs. First, we converted all images to grayscale. This step minimizes computational load while preserving critical structural layout markers. Second, we applied Contrast Limited Adaptive Histogram Equalization (CLAHE). This technique calibrates local contrast and emphasizes subtle pathological markers. Third, we utilized the Fast Non-Local Means Denoising algorithm to remove grain artifacts. This filter suppresses noise without blurring vital tissue boundaries.

Next, we normalized the pixel intensities by scaling all values to a fixed range between 0 and 1. We then resized each image to uniform dimensions of 224×224 pixels to match standard convolutional network inputs. Finally, we duplicated the single-channel arrays back into a three-channel configuration. This step ensures compatibility with pre-trained neural networks. Table 1 summarizes the parameters used during this stage.

3.3. Data Augmentation and Split

We applied mild augmentation techniques to every disease class to increase sample variety and prevent overfitting. These techniques adjusted image saturation levels and applied uniform lighting adjustments. We executed these variations using the Albumentations library. Table 2 outlines the specific parameters selected for this process.

Following augmentation, we randomly split the data into training (75%), validation (10%), and testing (15%) partitions. We applied stratified partitioning to maintain balanced class distributions across all subsets. We used a fixed random seed of 42 to ensure exact reproducibility. Table 3 presents the final image counts per subset for each

Table 2
Augmentation Techniques and Parameters

Technique	Parameters
Horizontal Flip	Prob. = 0.5
ShiftScaleRotate	shift_limit=0.03, scale_limit=0.05, rotate_limit=10°, p=0.5
RandomBrightnessContrast	brightness_limit=0.1, contrast_limit=0.1, p=0.5

Table 3
Dataset Distribution after Augmentation

Class	Total	Train (75%)	Val (10%)	Test (15%)
Eyelid	1575	1181	157	237
Normal	1947	1460	194	293
Cataract	1632	1224	163	245
Uveitis	669	501	66	102
Conjunctivitis	1071	803	107	161

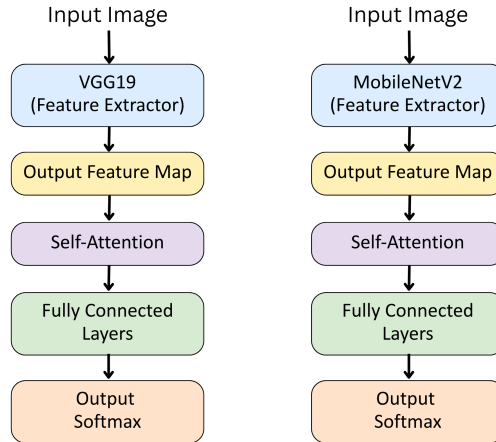


Figure 3: Architecture of Self Attention enhance VGG19 AND MobileNetV2.

disease category. For experimental evaluation, we encoded the labels into integers where Eyelid = 0, Normal = 1, Cataract = 2, Uveitis = 3, and Conjunctivitis = 4.

3.4. Baseline Models

We selected two standard convolutional neural network architectures as our experimental baselines: VGG19 and MobileNetV2. We modified both networks by inserting a self-attention mechanism. This mechanism allows the models to track long-range spatial relationships and focus on prominent feature maps within the eye images. The self-attention layer forces the networks to prioritize diseased tissue areas while disregarding irrelevant spatial background information. This property is crucial for detecting subtle lesion patterns. Figure 3 illustrates the structural composition of our self-attention-enhanced baseline networks.

3.5. Proposed Model: EyeDiseaseNet

We introduce a hybrid deep learning model named EyeDiseaseNet. This network fuses the deep feature representations extracted from our enhanced VGG19 and MobileNetV2 streams. This combination merges the complementary spatial details learned by both backbones. It leverages their individual structural advantages to assemble a robust

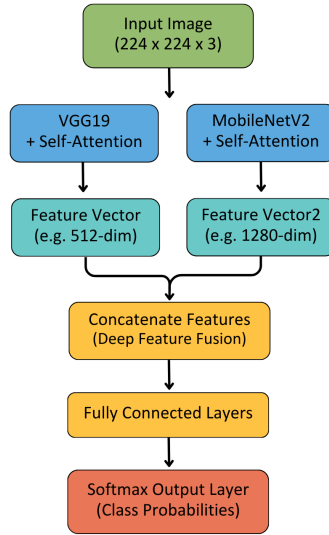


Figure 4: Architecture of proposed EyeDiseaseNet.

Table 4
Classification Metrics for VGG19

Class	Precision	Recall	F1-score
Cataract	0.9839	0.9959	0.9899
Conjunctivitis	0.9812	0.9752	0.9782
Eyelid	0.9873	0.9831	0.9852
Normal	0.9899	1.0000	0.9949
Uveitis	0.9694	0.9314	0.9500
Accuracy	0.9846		
Cohen's Kappa	0.9802		

feature representation block. The concatenated feature vector passes into deeply stacked fully connected layers for final classification. This multi-stream fusion strategy improves overall diagnostic accuracy by combining diverse feature spaces. Figure 4 displays the complete schematic diagram of EyeDiseaseNet.

3.6. Explainable AI

We integrated Gradient-weighted Class Activation Mapping (Grad-CAM) to establish model transparency and clinical interpretability. Grad-CAM tracks the gradient flow entering the final convolutional layers. It highlights the exact spatial zones in the image that determined the final diagnostic prediction. This approach generates clear visual evidence. This visualization helps clinicians inspect, verify, and trust the decisions made by EyeDiseaseNet. ““latex

4. Results and Discussion

We evaluated our proposed EyeDiseaseNet model alongside two baseline models: VGG19 and MobileNetV2. We computed standard classification metrics to assess performance across all five ocular categories. These metrics include precision, recall, F1-score, overall accuracy, and Cohen’s Kappa.

Table 4 displays the classification results for the baseline VGG19 model. VGG19 achieved an overall accuracy of 98.46% and a Cohen’s Kappa of 0.9802. These scores demonstrate a strong agreement with expert labels. The model achieved high precision and recall values for the Cataract and Normal classes. However, the recall score for Uveitis dropped to 0.9314. This drop suggests that the standard VGG19 architecture struggles to detect subtle features of low-contrast conditions. Overall, VGG19 serves as a solid baseline but leaves room for improvement.

Table 5
Classification Metrics for MobileNetV2

Class	Precision	Recall	F1-score
Cataract	0.9799	0.9959	0.9879
Conjunctivitis	0.9290	0.9752	0.9515
Eyelid	0.9868	0.9451	0.9655
Normal	1.0000	1.0000	1.0000
Uveitis	0.8800	0.8627	0.8713
Accuracy		0.9692	
Cohen's Kappa		0.9604	

Table 6
Classification Metrics for EyeDiseaseNet

Class	Precision	Recall	F1-score
Cataract	0.9917	0.9755	0.9835
Conjunctivitis	0.9758	1.0000	0.9877
Eyelid	0.9958	0.9916	0.9937
Normal	1.0000	1.0000	1.0000
Uveitis	0.9806	0.9902	0.9854
Accuracy		0.9913	
Cohen's Kappa		0.9889	

Table 5 outlines the performance metrics for the MobileNetV2 baseline. MobileNetV2 successfully identified all healthy eyes, reaching perfect scores across the Normal class. The network also performed well on Cataract detection. However, we observed a noticeable performance decline in more complex categories. Specifically, the recall for Uveitis dropped to 0.8627, and the precision for Conjunctivitis dropped to 0.9290. These results indicate that while MobileNetV2 is computationally efficient, it lacks the depth to capture fine, overlapping clinical features. It achieved an overall accuracy of 96.92% and a Cohen’s Kappa of 0.9604, making it the weakest performer among the evaluated models.

Table 6 provides the performance metrics for our proposed EyeDiseaseNet framework. Our model outperformed both VGG19 and MobileNetV2 across nearly all evaluation categories. EyeDiseaseNet maintained a perfect score for the Normal class, ensuring the model avoids false positives for healthy patients. Furthermore, the framework showed exceptional handling of challenging classes. It achieved F1-scores above 0.98 for both Uveitis and Conjunctivitis. The final overall accuracy reached 99.13%, and Cohen’s Kappa reached 0.9889. These values reflect near-perfect alignment with professional diagnostic labels, proving that our framework is highly suitable for real-world clinical use.

4.1. Analysis of Model Convergence and Training Stability

We tracked the training behavior of EyeDiseaseNet to observe learning speed and model stability. Figure 5 displays the loss and accuracy metrics over successive training epochs. The loss curves demonstrate that the model converges rapidly. EyeDiseaseNet successfully captures key ocular features within the first five epochs, where both training and validation loss drop sharply.

Unlike many deep architectures that exhibit high validation volatility, our validation loss closely followed the training loss. This trend confirms that our preprocessing and attention mechanisms prevent overfitting. We triggered an early stopping mechanism at epoch 21 to finalize the model weights at peak performance. The accuracy curves mirror this stability, rising quickly and flattening out above 98%. The minimal gap between the final training and validation curves confirms that the model generalizes well to new, unseen test data.

4.2. Robustness Across Thresholds: ROC and PR Analysis

We used Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves to confirm model robustness. In automated medical diagnosis, high ROC Area Under the Curve (ROC-AUC) scores are standard benchmarks.

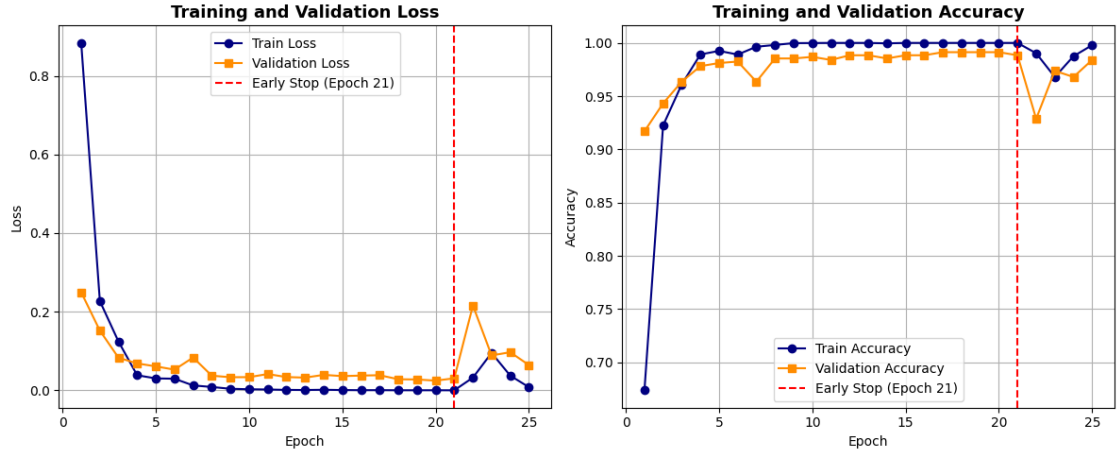


Figure 5: Training and validation learning curves of EyeDiseaseNet

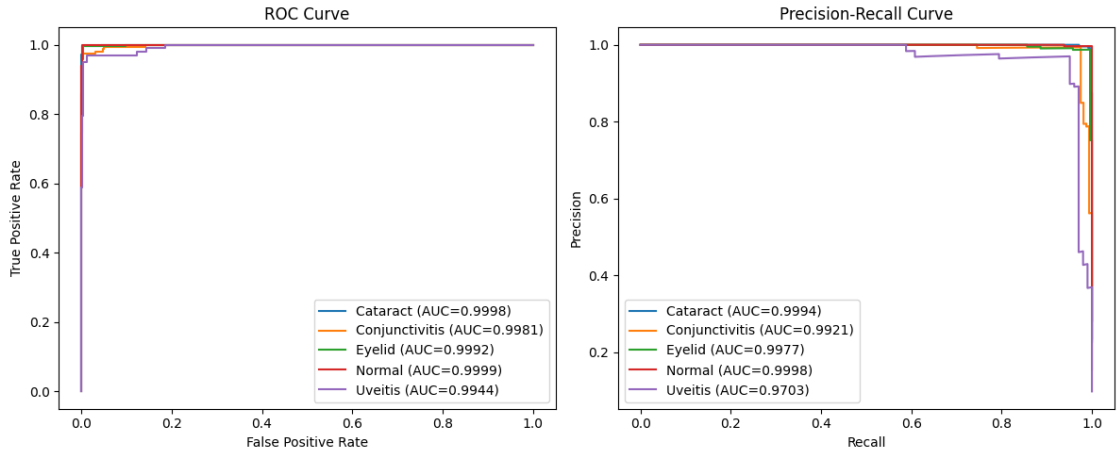


Figure 6: ROC and precision-recall curves demonstrating EyeDiseaseNet's performance and convergence.

However, Precision-Recall Area Under the Curve (PR-AUC) scores provide a stricter evaluation, particularly when class distributions vary.

As illustrated in Figure 6, EyeDiseaseNet achieved near-perfect AUC scores ranging between 0.9992 and 1.0000. The ROC curves align closely with the upper-left boundary, which indicates that our system maintains high sensitivity without increasing the false-positive rate. More importantly, the PR curves yielded scores between 0.9991 and 1.0000. These values confirm that positive predictions are highly accurate. This performance is crucial for ensuring clinical safety and building user trust.

4.3. Class-Specific Performance and Error Analysis

We evaluated a confusion matrix to observe how well EyeDiseaseNet differentiates between specific ocular pathologies. Figure 7 shows that the final predictions concentrate heavily along the main diagonal, confirming highly accurate classifications across all groups.

Notably, EyeDiseaseNet correctly categorized all 293 healthy eye samples. In clinical screening, avoiding false positives ensures that healthy individuals are not mistakenly referred for specialized care. We observed minor confusion between the Cataract and Conjunctivitis categories, which accounted for only four misclassified cases. This minor overlap is clinically understandable because both conditions can cause surface clouding and altered light reflections in the eye. For the Uveitis class, which is historically difficult to identify in small datasets, our model successfully

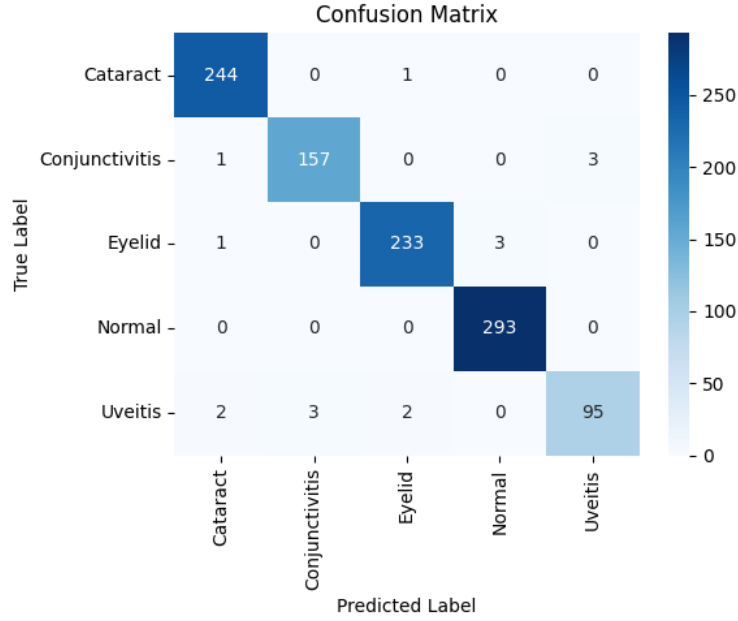


Figure 7: Confusion matrix of EyeDiseaseNet.

identified 101 out of 102 cases. Out of 1,036 total test images, the system made only 9 errors, demonstrating its excellent classification performance.

4.4. Computation Cost Analysis

We conducted an analysis of computational requirements to evaluate the practical efficiency of EyeDiseaseNet. As shown in Figure 8, EyeDiseaseNet required 554.46 seconds for total training. This requirement is much lower than the training times for baseline VGG19 (1057.26 seconds) and MobileNetV2 (1290.89 seconds). Our system required 12.14 seconds for test inference. Although this inference time is slightly higher than the standalone baselines, it remains well within acceptable parameters for real-time diagnostic use. These measurements show that our feature-fusion method achieves an optimal balance between accuracy and computational speed.

4.5. Interpretability of EyeDiseaseNet Using Grad-CAM

We integrated Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the regions that influenced our model's predictions. This step provides explainability by showing exactly where the network focuses its attention during evaluation. Figure 9 illustrates that the highlighted attention zones correspond closely with actual disease lesions on the eye surface. Our analysis confirmed that EyeDiseaseNet ignores background elements and consistently targets clinically relevant tissues. This transparency helps medical professionals verify model choices and can increase confidence in automated diagnostics.

4.6. Comparative Analysis of Existing Methods

Many recent studies have explored deep learning for eye condition screening. Most frameworks use variations of convolutional neural networks (CNNs) for classification tasks. However, many current architectures suffer from low accuracy or operate without explainability mechanisms. This lack of transparency limits their usefulness in real hospitals.

Table 7 provides a comparison between EyeDiseaseNet and recent state-of-the-art methods. Several existing systems focus on a single disease type or do not provide explainable AI outputs. In contrast, our framework covers five distinct eye conditions and integrates Grad-CAM for visual transparency. EyeDiseaseNet achieves a high diagnostic accuracy of 99.13% while keeping computational demands low. This balance makes our approach highly viable for clinical environments.

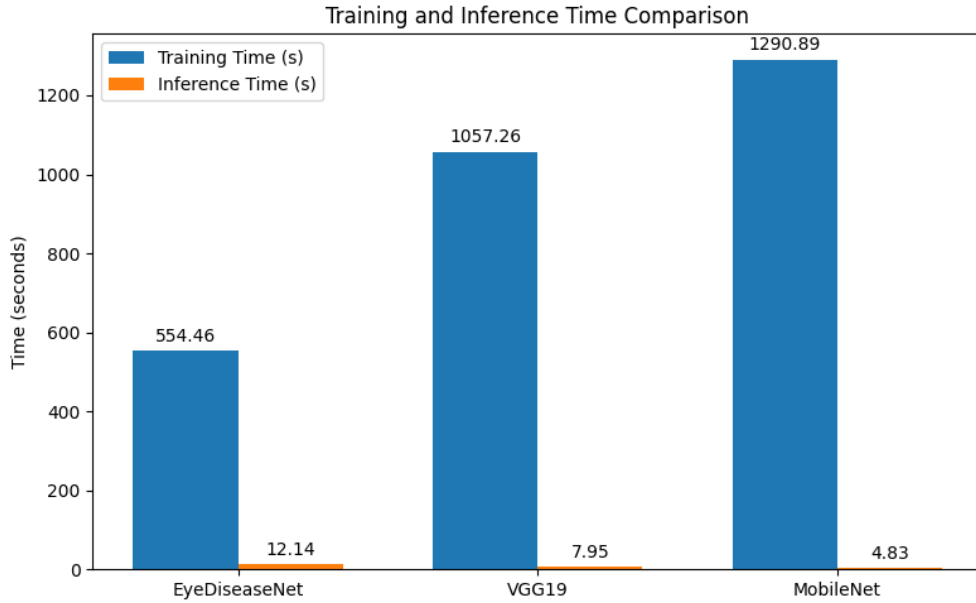


Figure 8: Computational Cost analysis of all evaluated models.

Table 7

Comparative analysis of existing models for eye disease detection

Study	Model/Method	Accuracy	XAI Used
Erдин et al. [8]	Custom CNN	92.1%	No
Rashid et al. [15]	Deep CNN (Multiclass)	94.6%	No
Nguyen et al. [12]	Hybrid CNN + Image Quadrature	96.2%	No
Zannah et al. [21]	Bayesian Optimized ML	95.3%	No
Lahari et al. [10]	CSDNet (Custom CNN)	97.1%	No
Umamaheswari et al. [19]	MobileNetV2-based CNN	91.7%	No
Ghadiri et al. [9]	Shallow Neural Network	90.2%	No
AbdElGhany et al. [1]	2D CNN + Grad-CAM	95.5%	Yes (Grad-CAM)
This Study	EyeDiseaseNet	99.13%	Yes (Grad-CAM)

Table 8

Feature-wise comparison between existing studies and EyeDiseaseNet

Feature	Existing Studies	EyeDiseaseNet
Multi-class Classification	Partial	✓
Feature Fusion	Rare	✓
Self-Attention	Limited	✓
Explainability (Grad-CAM)	Partial	✓
Computational Analysis	Rare	✓
Five Disease Categories	Rare	✓

4.7. Feature-wise Comparative Analysis

We also performed a architectural feature comparison to highlight the systemic advantages of EyeDiseaseNet. As outlined in Table 8, we evaluated several design traits, including multiclass capability, feature fusion, self-attention modules, and transparency features.

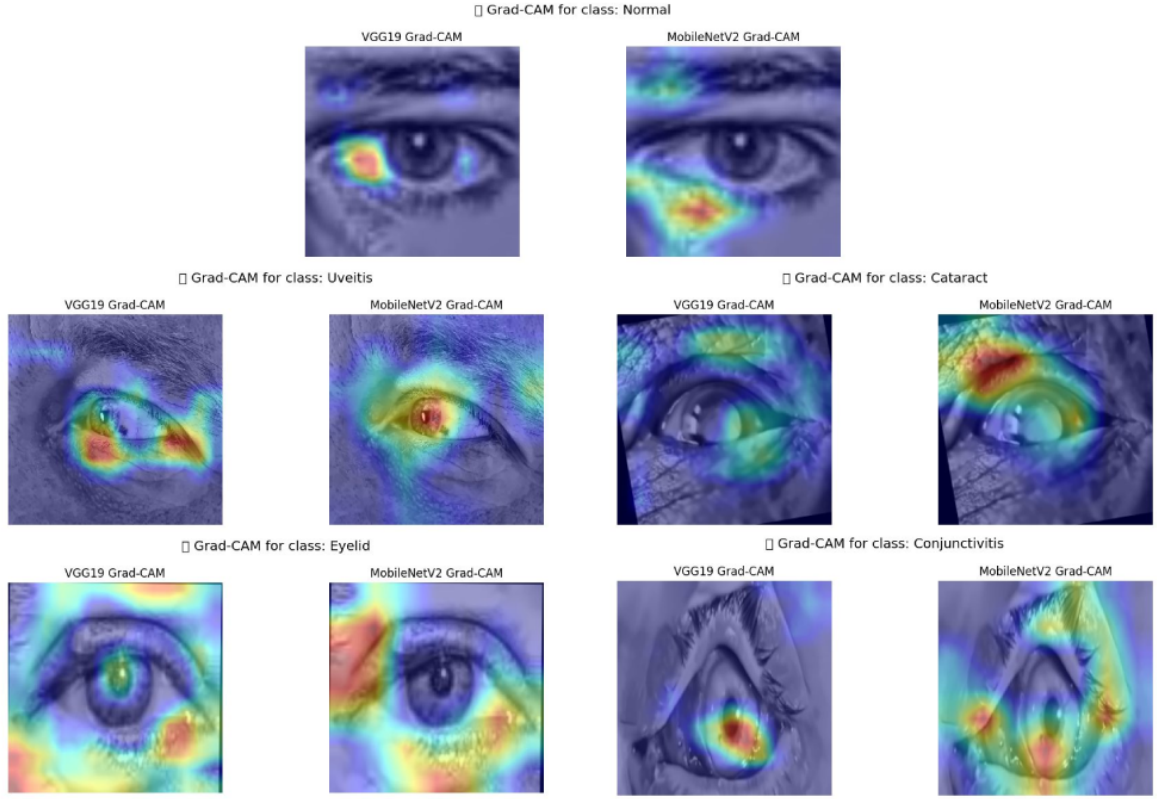


Figure 9: Grad-CAM visualizations of VGG19 and MobileNetV2 showing model attention over disease-affected areas.

The feature comparison in Table 8 indicates that EyeDiseaseNet includes all the essential components for a comprehensive screening platform. Most existing studies lack one or more structural features. They especially lag in multi-disease grouping and explainable outputs. This design makes EyeDiseaseNet a more practical and effective option for clinical deployment.

4.8. Practical Advantages of EyeDiseaseNet

EyeDiseaseNet offers several key advantages over current diagnostic systems. First, it identifies five distinct conditions within a single pipeline. This capability provides a comprehensive screening workflow for busy clinics. Second, our feature fusion method allows the model to separate visually similar eye conditions accurately. This reduces diagnostic errors. Third, the addition of Grad-CAM heatmaps removes the black-box nature of standard deep learning. It gives physicians a reliable way to verify predictions. These benefits make EyeDiseaseNet a reliable, accurate, and efficient tool for modern clinical decision support systems.

5. Conclusion

In this work, we developed EyeDiseaseNet to address the performance and transparency limitations of current ocular screening models. Our system integrates a feature-fusion network combining VGG19 and MobileNetV2 with a self-attention mechanism. This architecture improves overall feature representation. It achieves an overall classification accuracy of 99.13% and a Cohen's Kappa of 0.9889. These scores indicate near-perfect alignment with professional medical labels.

Our framework successfully handles visually similar conditions, such as Uveitis and Conjunctivitis, reaching F1-scores above 0.98 for both categories. This reliability is vital for early diagnostic accuracy. Furthermore, our structured

preprocessing pipeline uses CLAHE and Non-Local Means Denoising to optimize image quality and accelerate model convergence.

To resolve the black-box nature of standard deep learning networks, we integrated Grad-CAM to highlight the exact ocular regions driving each classification decision. This transparency is crucial for establishing clinical utility and building user trust. The model is computationally efficient, requiring 554.46 seconds for total training and 12.14 seconds for test inference. This balance makes our system highly suitable for real-time diagnostic workflows.

In future work, we plan to expand the diversity of our dataset to include prospective clinical evaluations. We also intend to design a lightweight web application to support real-time screening in low-resource environments. Ultimately, EyeDiseaseNet demonstrates that combining accuracy, explainability, and speed can improve automated ophthalmic diagnostics and broaden the use of deep learning in modern healthcare.

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Author contributions

S.D. performed data preprocessing, developed and implemented the proposed framework, conducted the experiments, analyzed the results, prepared the figures, and wrote the main manuscript. A.Q. supervised the research, provided continuous guidance and technical direction throughout the study, and critically reviewed and revised the manuscript. S.I. and A.S.S. contributed to data preprocessing, participated in technical discussions and decision-making, and assisted in manuscript writing and revision. All authors reviewed and approved the final manuscript.

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Data availability

The datasets analyzed during the current study are publicly available from their respective repositories. Additional processed data and implementation details are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study utilized publicly available retinal image datasets and did not involve direct interaction with human participants or the collection of personally identifiable information. Therefore, ethical approval and informed consent were not required.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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